CHAPTER 5

SYSTEM FUNCTION AND THE FREQUENCY RESPONSE

In our discussion on Fourier series and Fourier transforms we have seen that the system function of a LTI system is defined as the Fourier transform of the impulse response and plays a very important role in the analysis of such systems. In essence, the transfer function $H(\omega)$ is the "eigenvalue" associated with the "eigen function" $e^{j\omega t}$. Through another angle of view it is the "proportionality constant" relating the frequency domain representations of the input and the output. In this chapter we will investigate this important function and will try to develop a graphical representation for it.

5.1. THE POLAR REPRESENTATION: MAGNITUDE AND PHASE

In the preceeding chapters we have seen that the Fourier transform $X(\omega)$ of a time signal $x(t)$ is a complex function of ω in general. Such a function can be represented in rectangular coordinates interms of its real and imaginary parts.

Such a function can also be represented in polar coordinates interms of its magnitude and phase as

$$X(\omega) = |X(\omega)|.e^{j\angle X(\omega)} \quad . \tag{5.1.1}$$

As mentioned earlier, we can think of $X(\omega)$ as a decomposition of the signal $x(t)$ into a sum of periodic complex exponentials of different frequencies. Here $|X(\omega)|$ provides us with the information about the relative magnitudes of components at different frequencies. $\angle X(\omega)$, on the other hand, gives us the information concerning the relative phases of these components. Depending on what this phase function is, the resulting signals may look entirely different, as the following example demonstıates .

Example 5.1.1.

Consider the signal

$$z(t) = \sum_{k=-3}^{3} c_k.e^{jkt}$$

Where

$c_0 = 1,\ c_1 = c_1 = \frac{3}{4\sqrt{2}},\quad c_2 = c_{-2} = \frac{1}{2\sqrt{5}},\ \ c_3 = c_{-3} = \frac{1}{4\sqrt{10}}$.

The resulting signal is depicted in Fig.5.1.1 (a). A quick comparison of this signal with that of Example 3.2.2 reveals that both signals have Fourier series coefficients that are equal in magnitude, but here the phase angles are taken to be zero. The resulting signal shape is rather different from that of Example 3.2.2 which is shown in Fiğ.5.1.1 (b) for convenience.

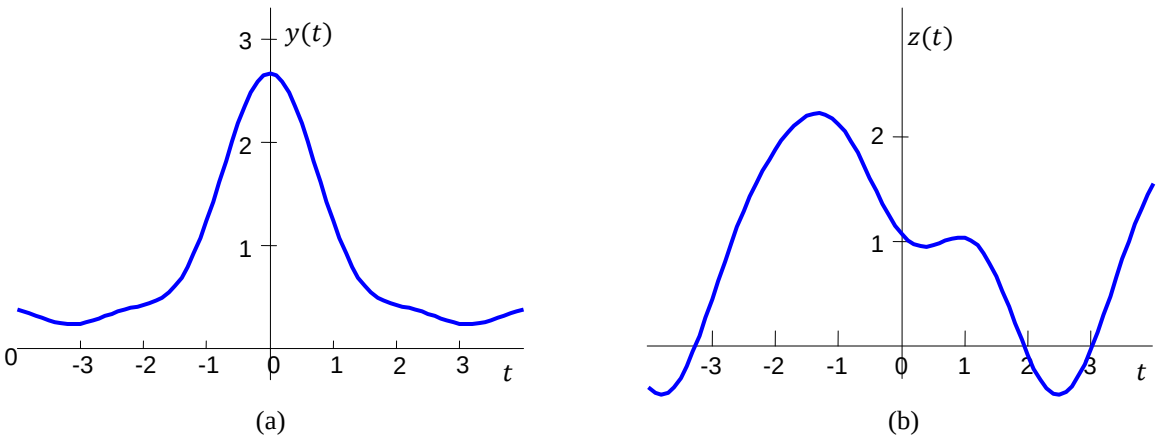

Figure 5.1.1 Example 5.1.1

The example above illustrates that changes in the phase of $X(w)$ results in severe changes in the corresponding time domain signal $x(t)$. In some instances "phase distortion" may be unimportant. For example it is often stated that human ear is insensitive to the phase of the sound. This statement is true to a certain extent, for if the phase of an individual spoken sound, (a vowel for example) is altered, the resulting signal will still be intelligible. However some severe changes in phase may result in loss of intelligibility. Consider, for example, the signal $x(-t)$ instead of $x(t)$. We know that

$$\mathcal{F}\{x(-t)] = X(-\omega) = X^*(\omega) \tag{5.1.2}$$

when $x(t)$ is real. Thus the Fourier transforms of $x(t)$ and $x(-t)$ have the same magnitude (i.e. $|X(\omega)| = |X(-\omega)|$) but they have opposite phases $(\angle X(\omega)) = -\angle X(-\omega)$. If $x(t)$ represents the tape recording of a speech signal, $x(-t)$ corresponds to playing it backwards. That is the spectrum of a tape recorded signal and the spectrum of the same signal in reverse have the same magnitude and differ only in phase. Obviously, this change in phase has a significant effect on the intelligibility of the speech signal.

5.2. BODE PLOTS

As we have seen in section 4.5.8 the input and output of a LTI system with the frequency response $H(\omega)$ are related by

$$Y(\omega) = H(\omega).X(\omega) \tag{5.2.1}$$

or in terms of the polar representation

$$|Y(\omega)| = |H(\omega)|.|X(\omega)| \tag{5.2.2a}$$

And

$$\angle Y(\omega) = \angle H(\omega) + \angle X(\omega) \quad . \tag{5.2.2b}$$

The magnitude of the frequency response is often referred to as the "gain" of the system because of the multiplicative form of (5.2.2a).

It is often convenient to represent the Fourier transforms graphically. To do this the magnitude and the phase can be plotted against ω . Although this is a useful and valid representation, (5.2.2a,b) suggest a modification. Note that if we have $\angle X(\omega)$ and $\angle H(\omega)$ plotted against ω, $\angle Y(\omega)$ can be obtained by simple addition. Similarly if $\log|H(\omega)|$ and $\log|X(\omega)|$ are plotted against ω , $\log|Y(\omega)|$ can be obtained easily by adding the corresponding values on these two graphs. For example, since the frequency response of the cascade of two LTI systems is the product of the individual frequency responses, the log-magnitude and the phase of the combination can be obtained by adding the individual log-magnitude and phase characteristics.

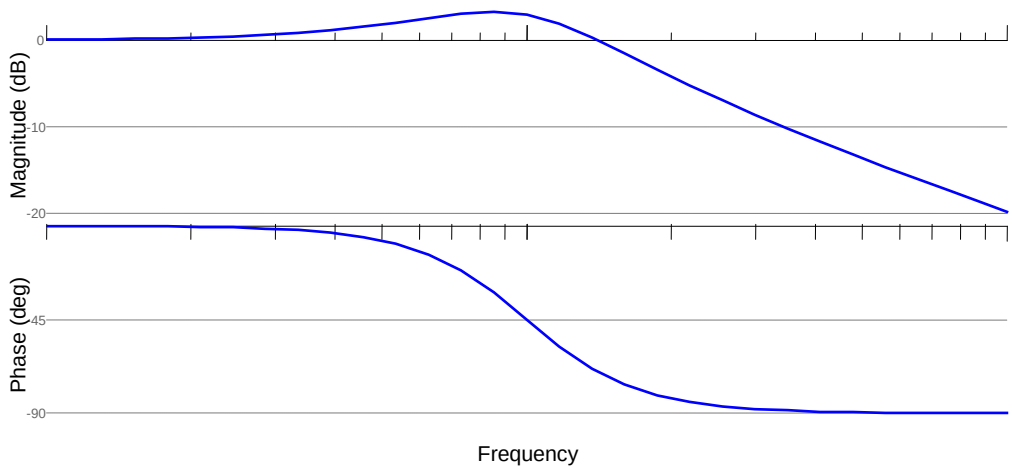

Figure 5.2.1. Typical Bode plots (Note the logarithmic scale)

The most widely known graphical representation is the "Bode plot" where $\angle H(\omega)$ and $20\log_{10}|H(\omega)|$ are plotted against ω . The quantity $20\log_{10}|H(\omega)|$ is the magnitude expressed in "decibels" (dB for short). Here 0 dB corresponds to $|H(\omega)| = 1$. Typical Bode plots are shown in Fig.5.2.1. In these plots usually a logarithmic scale is used for ω. In addition to yielding a better resolution when the frequency range of interest is large, this

choice also provides us with a very powerful method to obtain Bode plots for a given $H(\omega)$, as will be discussed in section 5.4. It is also worth noting that, Bode plots are shown for positive ω only. As we know, when $h(t)$ is real, $|H(\omega)|$ is an even function of ω, whereas $\angle H(\omega)$ is an odd function.

5.3. THE FREQUENCY RESPONSE FROM DIFFERENTIAL BQUATIONS

As we have discussed earlier, a particularly important class of continuous-time LTI systems are described by a constant-coefficient differential equation of the form

$$\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} a_k \frac{d^k x(t)}{dt^k} \quad . \tag{5.3.1}$$

The frequency response of such a system could be determined by first determining the impulse response using time domain techniques and then finding its Fourier transform. However, using the properties of the Fourier transform, the frequency response can be determined more easily. Assuming that Fourier transforms of $x(t)$and $y(t)$ exist, we can take the Fourier transform of (5.3.1), i.e.

$$\mathcal{F}\left\{\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k}\right\} = \mathcal{F}\left\{\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k}\right\} \quad . \tag{5.3.2}$$

Using the linearity pıoperty and the differentiation property, this becomes

$$\left\{\sum_{k=0}^{N} a_k (j\omega)^k\right\} Y(\omega) = \left\{\sum_{k=0}^{M} b_k (j\omega)^k\right\} X(\omega) \quad . \tag{5.3.3}$$

Since the frequency response is defined by the relation

$$Y(\omega) = H(\omega)X(\omega) \quad , \tag{5.3.4}$$

it is determined as

$$H(\omega) = \frac{Y(\omega)}{X(\omega)} = \frac{\{\sum_{k=0}^{M} b_k (j\omega)^k\}}{\{\sum_{k=0}^{N} a_k (j\omega)^k\}} \quad . \tag{5.3.5}$$

From (5.3.5) we see that $H(\omega)$ is a rational function in $j\omega$. The coefficients of the numerator are those of the right hand side and the coefficients of the denominator are those of the left hand side. Thus, the frequency response of a LTI system described by a differential equation can be written simply by inspection.

Example 5.3.1.

Consider the LTI system, which is initially at rest and is characterized by the differential equation

$$\frac{d^2 y(t)}{dt^2} + 3\,\frac{dy(t)}{dt} + 2y(t) = \frac{dx(t)}{dt} + 3x(t) \ .$$

The frequency response for this system is readily obtained from (5.3.5) as

$$H(\omega) = \frac{j\omega + 3}{(j\omega)^2 + 3j\omega + 2} = \frac{j\omega + 3}{(j\omega + 1)(j\omega + 2)} \quad .$$

To determine the corresponding impulse response, $H(\omega)$ is expressed as the sum of simpler terms, using "partial fraction expansion", as

$$H(\omega) = \frac{j\omega + 3}{(j\omega)^2 + 3j\omega + 2} = \frac{2}{j\omega + 1} + \frac{(-1)}{j\omega + 2} \quad .$$

Thus the impulse response is

$$h(t) = 2e^{-t}u(t) - e^{-2t}u(t) \ .$$

Example 5.3.2.

Let us determine the response of the system in Example 5.3.1 to the input

$$x(t) = e^{-t}u(t).$$

Then since

$$X(\omega) = \frac{1}{j\omega + 1} \ ,$$

transform of the output is

$$Y(\omega) = H(\omega)X(\omega) = \frac{j\omega + 3}{(j\omega + 1)^2(j\omega + 2)} \ .$$

The partial fraction expansion yields

$$Y(\omega) = \frac{(-1)}{j\omega + 1} + \frac{2}{(j\omega + 1)^2} + \frac{1}{j\omega + 2}$$

The inverse Fourier transform of the first and third terms can be determined easily like in the preceding examples. The inversion of the second term however requires either a more detailed table of transform pairs or use of some of the properties. Remembering the frequency differentiation property (4.5.3), i.e.

$$-jt.x(t) \leftrightarrow \frac{dX(\omega)}{d\omega}$$

and noting the fact that

$$\frac{d}{d\omega}\frac{1}{j\omega + 1} = -\frac{j}{(j\omega + 1)^2}$$

we obtain the transform pair

$$te^{-t} \leftrightarrow \frac{1}{(j\omega + 1)^2} \ .$$

Therefore,

$$y(t) = \{-e^{-t} + 2te^{-t} + e^{-2t}\}u(t) \, .$$

5.4. BODE PLOTS FOR FIRST AND SECOND ORDER FACTORS

We have seen that the frequency response of a LTI system characterized by a linear constant-coefficient differential equation is a rational function in $j\omega$.

Such a function can be expressed in general form

$$H(\omega) = K\frac{\prod_{k=1}^{M}(j\omega + z_k)}{\prod_{k=1}^{N}(j\omega + p_k)} \tag{5.4.1}$$

by factoring the numerator and denominator into first-order terms. Here, some of the z_k and p_k may be complex; however, if the differential equation characterizing the system has real coefficients, any complex p_k, or z_k appear in complex-conjugate pairs. Since

$$(j\omega + \lambda)(j\omega + \lambda^*) = |\lambda|^2 + 2\mathcal{R}e\{\lambda\} + (j\omega)^2, \tag{5.4.2}$$

(5.4.1) can be written as a ratio of products of first and second order factors with real coefficients:

$$H(\omega) = K.\frac{\prod_{k=1}^{P}[(j\omega)^2 + a_k.j\omega + b_k].\prod_{k=1}^{M-2P}(j\omega + z_k)}{\prod_{k=1}^{Q}[(j\omega)^2 + \alpha_k.j\omega + \beta_k].\prod_{k=1}^{M-2P}(j\omega + p_k)} \tag{5.4.3}$$

where P and Q indicate the number of conjugate pairs in the numerator and denominator respectively. (5.4.3) implies that, a LTI system described as above can be viewed as a cascade connection of first and second order systems.

As mentioned in Section 5.2, the log-magnitude and phase versus frequency characteristics of a cascade combination can be obtained by adding the log-magnitude and phase versus frequency characteristics of the individual systems forming the combination. Therefore, it is convenient to consider the Bode plots of first and second order factors. Bode plots for higher order transfer functions can be obtained by first factoring the frequency response as in (5.4.3) and adding the Bode plots for the individual components.

5.4.1. First Order Factors

Typically, a first order differential equation is expressed as

$$\tau.\frac{dy(t)}{dt} + y(t) = x(t) \tag{5.4.4}$$

The corresponding frequency response is

$$H(\omega) = \frac{1}{j\omega\tau + 1} \tag{5.4.5}$$

and the impulse response is

$$h(t) = \frac{1}{\tau} e^{-t/\tau} u(t) \tag{5.4.6}$$

as shown in Fig.5.4.1. The parameter τ is called the time constant of the first order system and it determines how fast the system responds. As τ decreases, the impulse response will decay towards zero more sharply, or the step response will aim towards its final value more quickly.

In Fig.5.4.2, the Bode plots of (5.4.5), i.e. log-magnitude and phase angle of $H(\omega)$ versus ω are shown, using a logarithmic scale for ω. In this figure we can see another advantage of using a log-scale for ω: we can easily obtain an approximate Bode plot for a first order system.

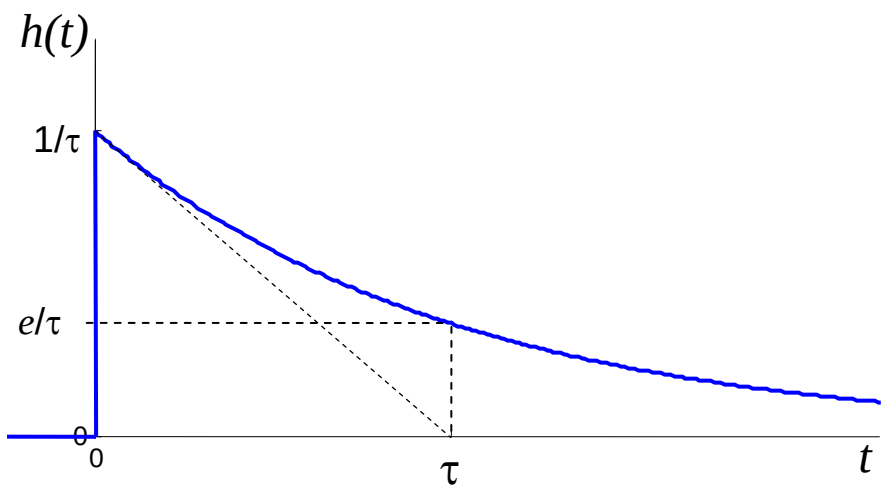

Figure 5.4.1. Impulse response of a first order system

Let us first consider the log-magnitude of the frequency response. From (5.4.5) we obtain

$$|H(\omega)|_{dB} = 20\log_{10}|H(\omega)| = 20\log_{10}[(\omega\tau)^2 + 1] \tag{5.4.7}$$

From (5.4.7) we see that

$$|H(\omega)|_{dB} = \begin{cases} 0 & \text{for } \omega\tau \ll 1 \\ -20\log\omega - 20\log\tau & \text{for } \omega\tau \gg 1 \end{cases} . \tag{5.4.8}$$

In other words, for a first order system, the low and high frequency asymptotes are straight lines, the former being the 0 dB line, and the latter being a straight line with a slope of “20 dB per decade" and crossing the 0 dB line at $\omega = 1/\tau$. This suggests a "piece-wise-straight-line" approximation to the Bode plot. This approximation is also shown in dashed lines in Fig.5.4.1. The frequency $\omega = 1/\tau$ where the approximation changes slope is called the "break frequency" or the "corner frequency". The break frequency is also the frequency where the difference between the actual log-magnitude and the straight line approximation reaches its maximum. The actual value of the log-magnitude at this point is

$$|H(1/\tau)|_{dB} = -10\log(2) \cong -3dB \ . \tag{5.4.9}$$

Thus an alternative name for the break frequency is the "3 dB"or "3 dB-down" frequency.

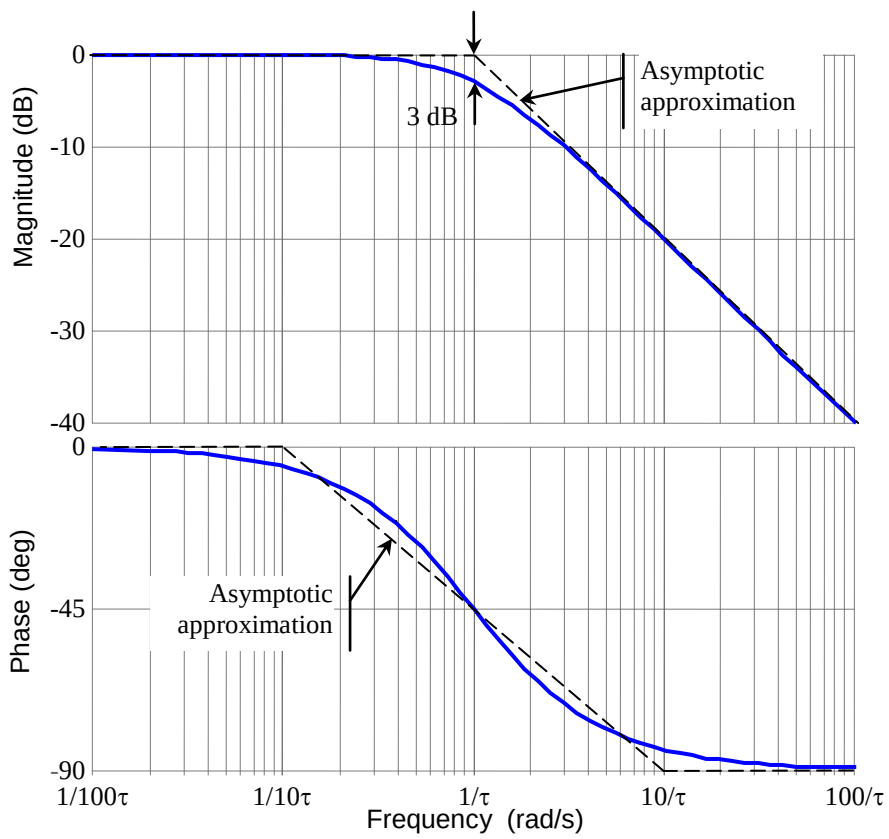

Figure 5.4.2. Bode plots for a first order system

It is also possible to obtain a satisfactory straight line approximation to $\sphericalangle H(\omega)$. From (5.4.5) the angle can be expressed as

$$\sphericalangle H(\omega) = -\tan^{-1}(\omega\tau). \tag{5.4.10}$$

Obviously

$$\sphericalangle H(\omega) = \begin{cases} 0 & \text{for } \omega\tau \ll 1 \\ -\frac{\pi}{2} & \text{for } \omega\tau \gg 1 \end{cases} \tag{5.4.11}$$

yielding the low and the high frequency asymptotes. However, these two asymptotes alone do not form a satisfactory straight-line approximation for the error becomes as high as $\pi/4$ around $\omega = 1/\tau$. These asymptotes, however, approximate the actual angle with an error less

0.2 radian outside the range $0.1 < \omega\tau < 10$, i.e., one decade below or above the corner frequency. If the two end points, i.e. 0 rad. at $\omega = 1/10\tau$ and $-\pi/2$ rad. at $\omega = 10/\tau$ are joined by a straight line, a satisfactory linear approximation will be obtained for the intermediate frequency range also. At the break frequency $\omega = 1/\tau$ this approximation will yield the actual value $\sphericalangle H(\omega) = -\pi/4$ and the maximum error will be at the two intersection points of the three asymptotes, the magnitude of the maximum erıor being less than 0.2 radian. The straight-line approximation for the angle plot is also shown in Fig.5.4.1 with dashed lines.

From the discussion above we can see once again, the inverse relationship between time and frequency domains. As we make τ smaller, $h(t)$ becomes more compressed toward the origin while $H(\omega)$ becomes broader since $|H(\omega)| \cong 1$ for a wider range of frequencies. This can also be observed from (5.4.5) and (5.4.6) since $h(t)$ is a function of t/τ and $H(\omega)$ is a function of $\omega\tau$.

5.4.2. Second Order Factors

A second order constant-coefficient differential equation of the type

$$\frac{d^2y(t)}{dt^2} + 2\zeta\omega_n\frac{dy(t)}{dt} + \omega_n^2 y(t) = \omega_n^2 x(t) \tag{5.4.12}$$

is often encountered in physical systems such as RLC circuits or mass-spring-damper type mechanical systems. The corresponding frequency response for such a diffeıential equation is

$$H(\omega) = \frac{\omega_n^2}{(j\omega)^2 + 2\zeta\omega_n(j\omega) + \omega_n^2} = \frac{1}{\left({}^{j\omega}/_{\omega_n}\right)^2 + 2\zeta\omega_n\left({}^{j\omega}/_{\omega_n}\right) + 1} \tag{5.4.13}$$

The denominator of this frequency response can be factored to yield

$$H(\omega) = \frac{\omega_n^2}{(j\omega - c_1)(j\omega - c_2)} \tag{5.4.14}$$

Where

$$c_{1,2} = -\zeta\omega_n \mp \omega_n\sqrt{\zeta^2 - 1} \quad . \tag{5.4.15}$$

For $\zeta \neq 1$, $c_1 \neq c_2$. and therefore, by partial fraction expansion we get

$$H(\omega) = \frac{M}{(j\omega - c_1)} - \frac{M}{(j\omega - c_2)} \tag{5.4.16}$$

Where

$$M = \frac{\omega_n}{2\sqrt{\zeta^2 - 1}} \quad . \tag{5.4.17}$$

Thus the corresponding impulse response is

$$h(t) = M[e^{c_1 t} - e^{c_2 t}]u(t) \quad . \tag{5.4.18}$$

For $\zeta = 1$, on the other hand, $c_1 = c_2$ and therefore

$$H(\omega) = \frac{\omega_n^2}{(j\omega + \omega_n)^2} \tag{5.4.19}$$

yielding

$$h(t) = \omega_n^2 t e^{-\omega_n t} u(t) \quad . \tag{5.4.20}$$

Note from (5.4.13), (5.4.18) and (5.4.20) that while $H(w)$ is a function of ω/ω_n, $h(t)$ is a function of $\omega_n t$. Thus changing ω_n is essentially equivalent to scaling the independent variables in time and frequency domains with the usual inverse relationship.

Fig.5.4.3. depicts $h(t)$ for different values of ζ. As one can easily notice, for $0 < \zeta < 1$, the impulse response shows oscillatory behaviour and the system is said to be underdamped. If $\zeta > 1$, both c_1 and c_2 are real and $h(t)$ consists of two decaying exponentials. In this case the system is said to be overdamped. The borderline case $\zeta = 1$, where $c_1 = c_2$ is what's so called the critically damped case.

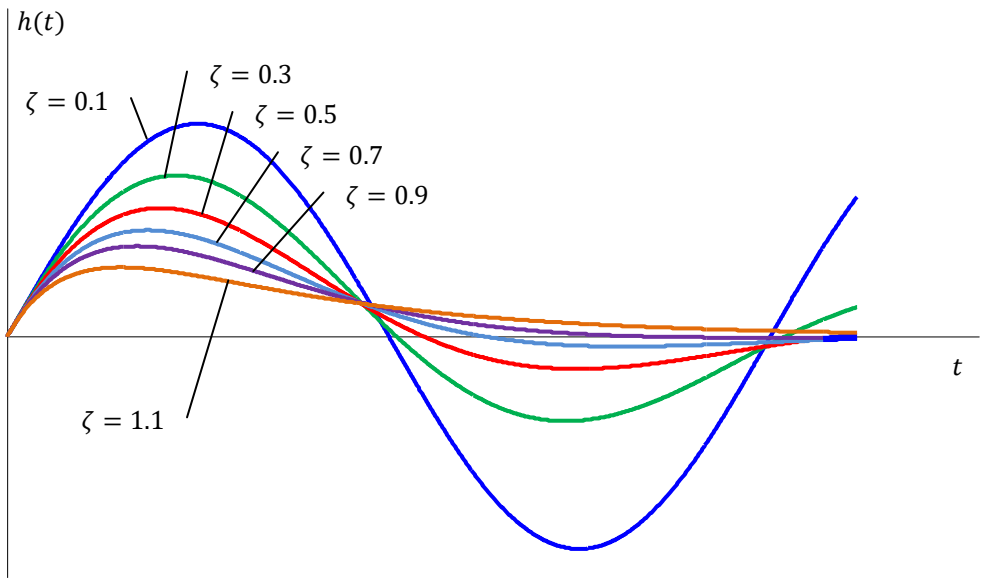

Figure 5.4.3. Impulse response of a second order system with several values of the damping ratio ζ

When ζ is reduced, the oscillations are sustained for a longer time and at the limit as $\zeta \to 0$, we obtain undamped oscillations of angular frequency ω_n. Thus the parameter ω_n is referred to as the "undamped natural frequency" whereas the parameter ζ is called the "damping ratio".

In Fig.5.4.4, the Bode plots of the frequency response given in (5.4.13) are shown for several values of the damping ratio. Again a logarithmic frequency scale is used. As in the first order case we have straight-line asymptotes for very low and very high frequencies. Specifically, from

$$\begin{aligned} |H(\omega)|_{dB} &= 20\ \log|H(\omega)| \\ &= -10\ \log\left\{\left[1-\left({}^{\omega}/_{\omega_n}\right)\right]^2 + 4\zeta^2\left({}^{\omega}/_{\omega_n}\right)^2\right\} \end{aligned} \tag{5.4.21}$$

öne can see that

$$|H(\omega)|_{dB} = \begin{cases} 0 & \text{for } \omega \ll \omega_n \\ -40\log(\omega) + 40\log(\omega_n) & \text{for } \omega \gg \omega_n \end{cases} . \tag{5.4.22}$$

Therefore the low frequency asymptote of the log-magnitude is the 0 dB line and the high frequency asymptote is a straight line crossing the 0 dB line at $\omega=\omega_n$ with a slope of -40 dB per decade. Thus we again have a straight-line approximation to the Bode plot consisting of two segments with a "break frequency" of $\omega=\omega_n$, as shown as a dashed line on Fig.5.4.4.

We can also obtain a straight-line approximation to $\sphericalangle H(\omega)$. The exact expression being

$$\sphericalangle H(\omega) = -\tan^{-1}\left\{\frac{2\zeta\left({}^{\omega}/_{\omega_n}\right)}{1-\left({}^{\omega}/_{\omega_n}\right)^2}\right\} \tag{5.4.23}$$

it can easily be seen that

$$\measuredangle H(\omega) = \begin{cases} 0 & \text{for } \omega \ll \omega_n \\ -\pi & \text{for } \omega \gg \omega_n \end{cases} . \tag{5.4.24}$$

These approximations will yield satisfactory results when $\omega \leq \omega_n/10$ or $\omega \geq 10\omega_n$. The straight line approximation in the intermediate range can be obtained by joining the points "0 rad. at $\omega = \omega_n/10$" and "$-\pi$ rad. at $\omega = 10\omega_n$". Note that this approximation yields the actual value at $\omega = \omega_n$, i.e. $\measuredangle H(\omega) = -\pi/2$.

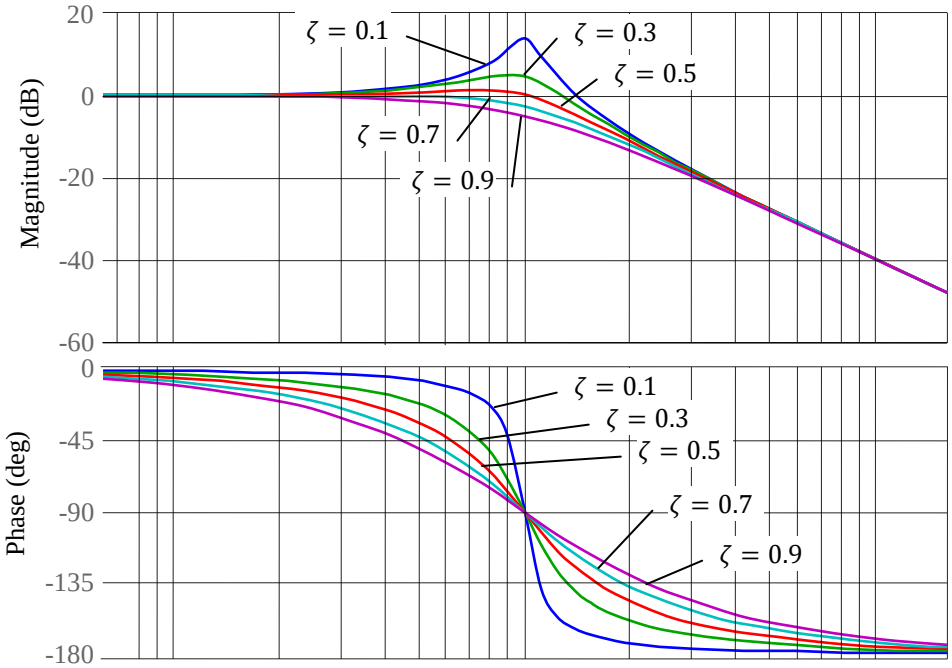

Figure 5.4.4 Bode plots for a second order system with several values of the damping ratio ζ

It is important to observe that although the straight-line approximations for both log-magnitude and angle plots do not depend on the damping factor, actual plots do. Hence the approximation error depends on ζ. For the log magnitude plot, this error is most pronounced in the vicinity of (although not exactly at) the break frequency. It can be shown by the reader that for $\zeta < 1/\sqrt{2}$, $H(\omega)$ will have a maxirnum at

$$\omega_{max} = \omega_n\sqrt{1-\zeta^2} \tag{5.4.25}$$

and the value of the maximum is

$$|H(\omega_{max})| = \frac{1}{2\zeta\sqrt{1-\zeta^2}} \tag{5.4.26}$$

The fact that $|H(\omega)|$ can have a peak is extremely useful especially when one desires to emphasize components of a certain frequency. Such an application is probably the simplest case of frequency selective filters. The selectivity of such a "filter" depends on the sharpness of the peak. The "quality factor" Q, which is usually taken as $Q = {}^{1}/_{2\zeta}$ for a system of the type (5.4.12) is a measure of this sharpness: As Q is higher, ζ is lower and thus the peak is sharper!

One can also observe that as ζ is reduced, the approximation error in the angle is also increased. Although for all ζ values, angle is still $-\pi/2$ at $\omega = \omega_n$, it changes rapidly in the vicinity and changes will be sharper for smaller ζ values.

We have indicated earlier that first-order and second-order factors can be used as building blocks for higher order frequency responses. The procedure to obtain the Bode plot for a more complex frequency response is demonstrated by the following example

Example 5.4. 1

Consider the frequency response

$$H(\omega) = \frac{1 + j\omega}{(1 + 10j\omega)(-\omega^2 + 5j\omega + 25)}$$

This frequency response can be viewed as a product of 4 terms as

$$H(\omega) = \frac{1}{25}.(1 + j\omega).\frac{1}{1 + 10j\omega}.\frac{1}{\left({}^{j\omega}/_{5}\right)^2 + 2.(0.5).\left({}^{j\omega}/_{5}\right) + 1}$$

Approximate and exact plots for each of these factors and the overall frequency response are shown in Fiğ.5.4.5.

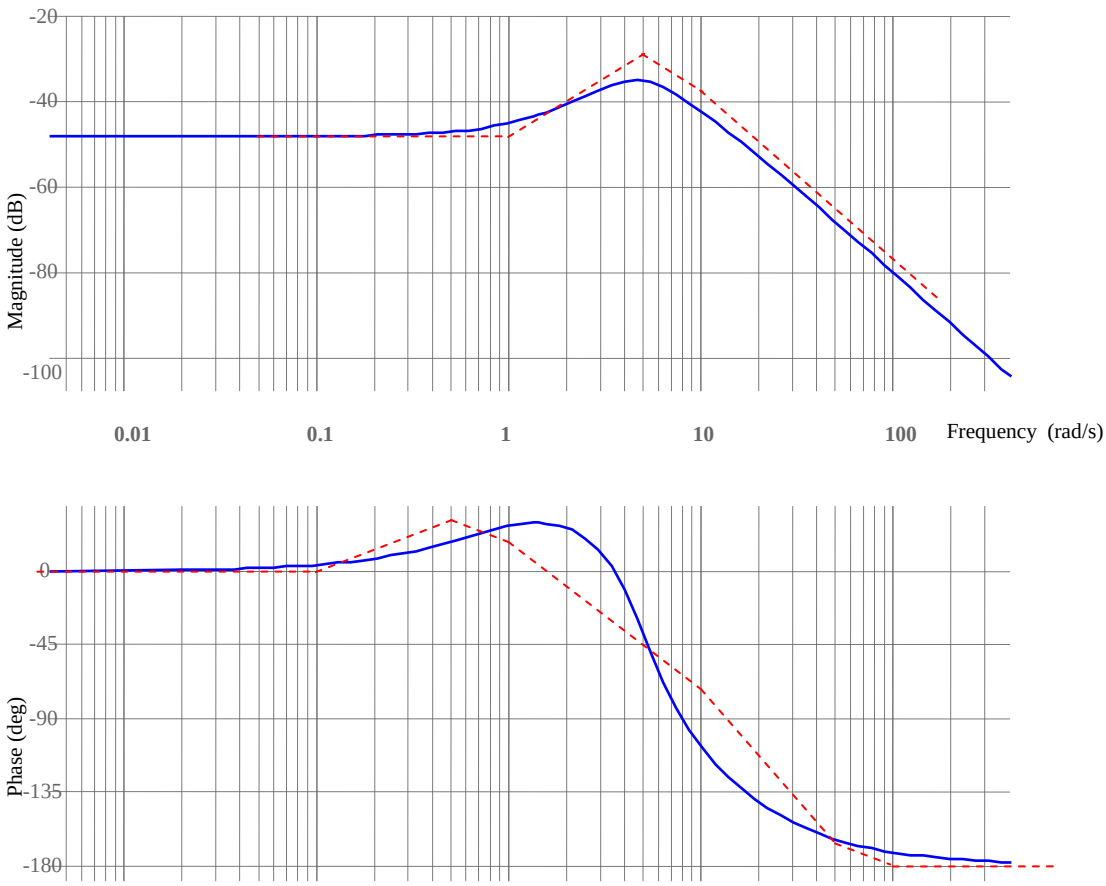

Figure 5.4.5. Example 5.4.1.

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PROBLEMS

Problem 1.

Sketch the magnitude-frequency response (Bode) plot for the linear, time-invariant system governed by the differential equation

$$\ddot{y}(t) - 4y(t) = \dot{x}(t) - 2x(t)\,.$$

Problem 2.

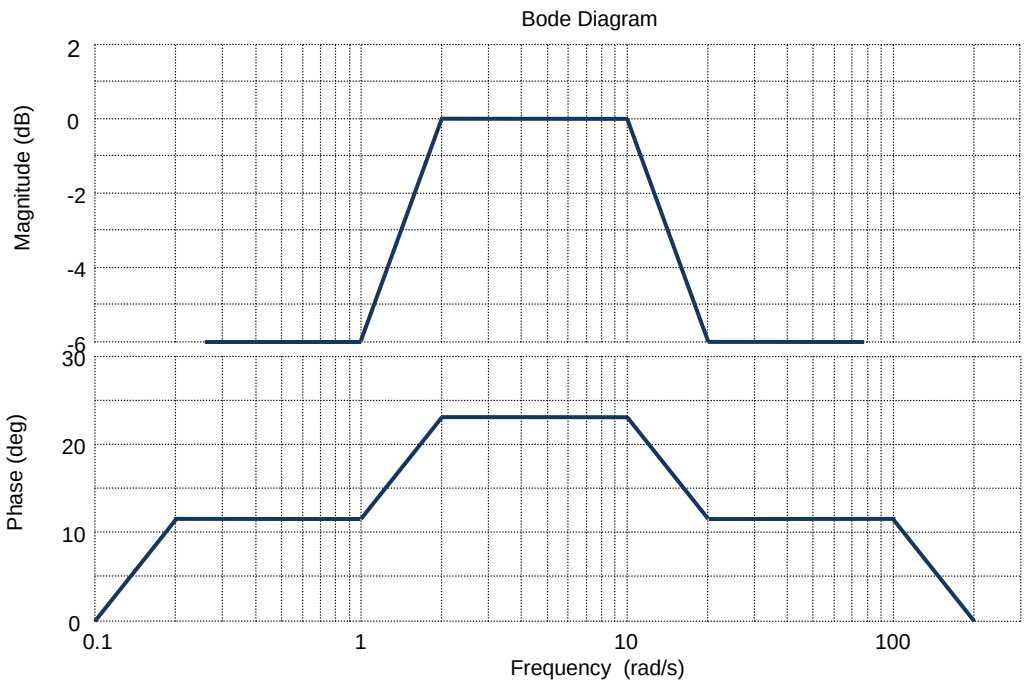

Figure shows the Bode (corner) plots for the magnitude and phase vs. frequency characteristics of a certain linear, time-invariant continuous-time system. Determine the system function $H(\omega)$.